SPACE NETRA

Satellite Imagery Intelligence & Change Detection Platform

The Consolidated 8-Part System & Feasibility Observation Report

Target Area of Interest:	Chhatrapati Sambhajinagar (Aurangabad), Maharashtra, India
Baseline Imagery Platform:	Copernicus Sentinel-2 Level-2A (10m Multispectral) via MS Planetary Computer STAC
Core Machine Learning:	OpenAI CLIP (ViT-B/32) + Facebook AI Similarity Search (FAISS IndexFlatIP)
Document Purpose:	Unified Architectural, Remote Sensing, Risk, and Engineering Roadmap Specification
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This document synthesizes all analytical sessions into an authoritative technical reference for project implementation.

OBSERVATION 1

Executive Project Understanding & Strategic Mission

Space Netra is conceived as a dual-capability geospatial intelligence engine developed for academic demonstration and regional remote sensing evaluation in Maharashtra, India. The application bridges modern foundation AI models with classical Earth observation methodologies to address two critical spatial challenges: unstructured visual discovery and deterministic land-surface change monitoring.

The Core Mission: Decoupled Dual Capabilities

- **Capability A (Semantic Spatial Search):** Enables natural-language text retrieval (e.g., 'urban area', 'reservoir', 'airport') over a satellite image archive using OpenAI CLIP (ViT-B/32) multi-modal embeddings and Facebook AI Similarity Search (FAISS).
- **Capability B (Multi-Temporal Change Detection):** Ingests calibrated satellite observations across two distinct historical timestamps (T1 vs. T2) over an Area of Interest (AOI) to identify, quantify, and map physical surface transitions.

Problem Statement & Target Beneficiaries

Earth observation constellations capture petabytes of raw pixels daily. Municipal authorities, environmental monitors, and spatial planners face an analytical bottleneck: manually cross-referencing multi-temporal scenes is labor-intensive, while querying imagery traditionally requires complex GIS coordinates rather than intuitive human descriptions.

User Group	Operational Goal	Platform Benefit
Urban & Municipal Planners	Monitor peri-urban growth & layouts	Rapidly flag newly leveled parcels and roadwa
Environmental Monitors	Track deforestation & water drying	Quantify surface water depletion across seaso
Defense & Spatial Analysts	Tactical spatial intelligence	Query archives using natural language without
Academic Researchers	Model benchmarking & evaluation	Compare zero-shot CLIP and baseline differenc

Search vs. Change Detection: The Fundamental Divide

A critical finding of this evaluation is that Image Similarity Search and Temporal Change Detection solve fundamentally different mathematical problems and must not share the same analytical pipeline. Search evaluates global semantic proximity in an unreferenced latent space, while Change Detection requires sub-pixel coordinate registration, identical sensor bandpasses, and physical surface reflectance calibration.

OBSERVATION 2

System Architecture & 10-Area Component Status Audit

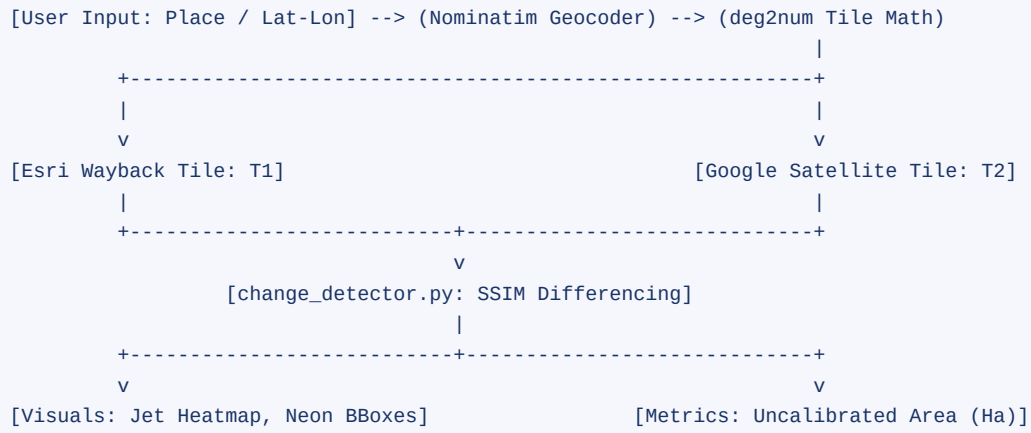
A rigorous line-by-line inspection of the active source files (app.py, change_detector.py, ingest.py, download_images.py) was conducted to separate verified functional code from documented claims and planned enhancements.

Component Verification Matrix

Component / Subsystem	Technical Implementation	Verification State
Application Architecture	Monolithic procedural Streamlit shell (app.py	Verified Existing
User Interface	Streamlit dual-mode selector, Folium maps, Pl	Verified Existing
Nominatim Geocoding	Resolves city text queries to Lat/Lon via geo	Verified Existing
Slippy Tile Fetcher	Synchronous HTTP GET scraping Google & Esri 2	Verified Existing (Defective)
CLIP Embedding Model	OpenAI ViT-B/32 generates 512-d unit-normaliz	Verified Existing
FAISS Vector DB	IndexFlatIP cosine similarity index (spatial.	Verified Existing
Change Detector Engine	OpenCV + skimage SSIM differencing on 8-bit g	Verified Existing (Uncalibrated)
Area Calculation Math	Multiplies changed pixels by arbitrary UI sli	Documented Unverified
STAC API Ingestion	Programmatic Sentinel-2 L2A query via Planeta	Planned / Proposed
Metric Reprojection	Rasterio warping to UTM Zone 43N (EPSG:32643)	Planned / Proposed

Data Flow Architecture of Current Implementation

CURRENT PROCEDURAL PIPELINE IN APP.PY



ARCHITECTURAL ALERT: FIVE CRITICAL WEAKNESSES IN ACTIVE CODE

1. Monolithic Coupling: Network calls, UI, and CV differencing execute in a single blocking thread. 2. Cross-Sensor Flaw: Comparing Esri Wayback with Google Satellite causes massive illumination false alarms. 3. Lack of Georeferencing: Scraped Web Mercator tiles lack affine transforms; displayed hectares are speculative. 4. Cache File Collisions: Hardcoded temp paths (temp_t1_historical.jpg) cause race conditions in concurrent sessions. 5. Terms of Service Risk: Direct scraping of Google Satellite tiles violates platform terms and risks IP blacklisting.

OBSERVATION 3

Remote Sensing & Computer Vision Feasibility Study

Satellite image change detection requires strict adherence to Earth observation physics. This section evaluates what can be scientifically achieved with optical satellite sensors versus common academic misconceptions.

Surface Change Detectability by Physical Scale

- **Macro Urban Expansion (High Feasibility):** Conversion of vegetation or farmland into industrial plots, packed soil, or paved corridors. At 10m Ground Sampling Distance (GSD), multi-hectare clearings generate distinct spectral transitions.
- **Vegetation Canopy Loss (Very High Feasibility):** Deforestation, crop harvesting, and land clearing exhibit drastic drops in Normalized Difference Vegetation Index (NDVI = (NIR - Red) / (NIR + Red)) using Sentinel-2 B4 and B8 bands.
- **Surface Water Dynamics (Very High Feasibility):** Reservoir depletion and lake surface fluctuations around Maharashtra are easily delineated via Normalized Difference Water Index (NDWI) due to near-total NIR absorption by water.
- **Individual Building Detection (Unfeasible at 10m):** A typical residential structure (100-200 m^2) occupies only 1 to 2 pixels on a 10m Sentinel-2 grid. Mixed-pixel point spread functions make resolving individual house construction physically impossible without sub-meter commercial imagery.

Classical Computer Vision vs. Supervised Deep Learning

A foundational strategic decision is choosing between classical spectral/structural differencing and deep neural networks (e.g., Siamese U-Net, ChangeFormer). The trade-offs for a student pilot are decisive:

Evaluation Dimension	Classical Differencing (SSIM / NDVI)	Supervised Deep Learning (CNN/Transformers)
Training Data Needed	Zero training samples required	Thousands of audited, paired change masks
Computational Overhead	Instantaneous CPU execution (< 1 sec)	Heavy GPU training & inference requirements
Explainability	100% transparent physical reflectance math	Black-box feature representation
Domain Adaptation	Globally consistent across all regions	Severe degradation outside training domain
Project Suitability	Mandatory baseline for initial release	High-risk scope trap; defer to post-pilot

Scientifically Unjustified Claims to Avoid

- **Claim 1 (Building Outlines):** Never claim detection of individual residential houses using 10m Sentinel-2 data.
- **Claim 2 (Real-Time Enforcement):** Never claim real-time monitoring; Sentinel-2 has a 5-day revisit cadence, frequently occluded by monsoonal cloud cover.
- **Claim 3 (Unsubstantiated Accuracy):** Never state '95% accuracy' without reporting Precision, Recall, and IoU against an audited test dataset.
- **Claim 4 (Tile-Based Hectares):** Never report physical area figures derived from uncalibrated Web Mercator slippy-map tiles.

OBSERVATION 4

Open Satellite Imagery Sources & Benchmark Datasets

A rigorous survey of Earth observation platforms, open satellite archives, and preprocessed academic datasets was conducted to determine the optimal data ingestion pipeline for Space Netra.

Comprehensive Satellite Data Evaluation Matrix

Data Source / Platform	Data Modality	Resolution	Coverage / Revisit	Licensing & Cost	PoC Viability
Sentinel-2 L2A (MPC)	STAC API / COGs	10m Vis/NIR	Global / 5-day revisit	Free & Open Access	Rank 1 (Optimal)
Copernicus Data Space	STAC / OData API	10m Vis/NIR	Global / 5-day revisit	Free (EU Law)	Rank 2 (Fallback)
USGS Landsat 8/9	STAC / COGs	30m Optical	Global / 8-day dual sat	Public Domain	Macro Changes Only
OSCD Benchmark	Static Benchmark	10m Sentinel-2	24 Fixed Global Cities	Open Research	Model Validation Only
SpaceNet 7 Dataset	Static Time-Series	4.0m PlanetScope	100 Sites / 24 Months	CC BY-SA 4.0	Too Heavy for V1
Google/Esri Tiles (Current)	Rendered RGB Tiles	Uncalibrated	Ad-hoc snapshots	Scraping Violation	Unsound (Replace)

Why Microsoft Planetary Computer (MPC) STAC Wins

Microsoft Planetary Computer provides the premier infrastructure for Space Netra's change detection engine. It exposes calibrated Sentinel-2 Level-2A Bottom-of-Atmosphere (BOA) surface reflectance directly in Cloud-Optimized GeoTIFF (COG) format. Key engineering advantages include:

- **Windowed Range Reads:** Using rasterio, the application reads only the exact byte ranges corresponding to the 2x2 km Sambhajinagar AOI directly from cloud storage, downloading ~3-5 MB rather than 800 MB full-scene archives.
- **Per-Pixel SCL Quality Masking:** Provides a 20m Scene Classification Layer (SCL) that flags high/medium probability clouds, cirrus, and cloud shadows deterministically, allowing automated pixel masking.
- **Zero Cost & Python Integration:** Fully accessible via pystac-client and planetary-computer SDKs without complex enterprise authentication hurdles.

CRUCIAL DISTINCTION: REAL IMAGERY VS. BENCHMARK DATASETS

Datasets like OSCD and SpaceNet 7 are pre-packaged benchmark chips over fixed foreign cities (Paris, Las Vegas, etc.). They are invaluable for validating model accuracy in an academic report, but CANNOT be queried over Chhatrapati Sambhajinagar. Planetary Computer STAC is mandatory for monitoring custom real-world geographic coordinates.

OBSERVATION 5 Product Management Feature Evaluation & Prioritization

A strict technical product management audit was applied to 12 candidate features to balance scientific defensibility, student developer bandwidth, and system stability.

Feature Prioritization Matrix

Feature Description	User Value	Complexity	Misleading Risk	Classification
1. Bounded AOI Map Selector	High	Medium	Low	Essential (V1 Scope)
2. Two-Date Image Ingestion	Critical	Medium	Medium	Essential (V1 Scope)
3. Registered Image Differencing	Critical	Medium-High	High	Essential (V1 Scope)
4. Visual Change Map & Heatmap	High	Low-Medium	Medium	Essential (V1 Scope)
5. Detected Region Bounding Boxes	Medium-High	Medium	High	Useful After V1 (V2)
6. Area Percentage Change Metric	Medium	Very Low	Low-Medium	Essential (V1 Scope)
7. Ground Hectare Area Calculation	High	Medium	Extreme if Uncal.	Essential (V1 Scope)
8. Natural Language Text Search	High	Low (Done)	Low	Essential (Preserve CLIP)
9. Reverse Image Similarity Search	Medium	Low	Low	Useful After V1 (V2)
10. Subjective Significance Ranking	High	High	High	Not Recommended Yet
11. Cloud, Shadow & Seasonality Filter	Critical	High	Very High if Omit.	Essential (V1 Scope)
12. Provenance & Limitations Banner	Very High	Very Low	Eliminates Risk	Essential (V1 Scope)

Strategic Scope Inclusions and Exclusions

The rationales behind key product boundaries are defined as follows:

- **Include Feature 8 (CLIP + FAISS Search):** The image retrieval engine is already functional in ingest.py and app.py. Discarding it wastes working assets; isolating it in a dedicated tab delivers instant multi-modal functionality at zero added cost.
- **Include Feature 11 (Cloud Masking & Anniversary Dates):** If omitted, a single cloud over Sambhajinagar generates hundreds of hectares of fake construction alarms, instantly disqualifying the project in any technical review.
- **Include Feature 12 (Limitations Display):** Explicitly presenting sensor resolution constraints (10m) in the UI transforms an overambitious demo into a credible engineering pilot.
- **Exclude Feature 10 (Significance Ranking):** Without audited ground truth, ranking which change is 'more important' is arbitrary guesswork that cannot be defended.
- **Exclude Custom Deep Learning Training:** The 5 unlabelled images in data/raw/ cannot train a neural network. Attempting deep learning training in Week 1 guarantees project failure.

OBSERVATION 6

Technical Risk Audit & 13 Credibility Pitfalls

Thirteen critical remote sensing risks were audited to identify failure modes, evaluation protocols, and required evidence before deploying the change detection engine.

The 13 Remote Sensing Failure Modes & Mitigations

- **1. Solar Illumination Shift:** Solar zenith differences cast shifting shadows. Mitigation: Enforce anniversary date matching (+/- 20 days) to equalize solar elevation angles.
- **2. Seasonal Phenology:** Monsoon greening vs. summer dry-down mimics land clearing. Mitigation: Compare identical calendar months across years.
- **3. Cloud & Shadow Contamination:** Popcorn clouds trigger intense change spikes. Mitigation: Mask out Sentinel-2 SCL cloud and shadow classes prior to differencing.
- **4. Resolution (GSD) Mismatch:** Resampling 30m Landsat to 10m Sentinel-2 causes edge blur. Mitigation: Enforce strict single-sensor pairings.
- **5. Spatial Misalignment Jitter:** A 0.5-pixel shift creates massive edge halos. Mitigation: Harmonize rasters to identical UTM 43N grid specs using rasterio.
- **6. Cross-Sensor Discrepancy:** Comparing Google tiles with Esri tiles proves nothing. Mitigation: Pure Sentinel-2 L2A to Sentinel-2 L2A comparisons.
- **7. Building Detection Physical Limits:** 10m GSD cannot resolve small houses. Mitigation: Rescope claims strictly to macro land transformation and industrial layouts.
- **8. Digital Compression Artifacts:** Lossy JPEG web tiles introduce 8x8 block noise. Mitigation: Ingest native uncompressed GeoTIFF rasters directly.
- **9. Invalid Area Mathematics:** Web Mercator distortion overstates area by ~1,900%. Mitigation: Project to UTM Zone 43N (EPSG:32643) where 1 px = 0.01 ha.
- **10. Historical Imagery Gaps:** Arbitrary date pickers fail on cloudy days. Mitigation: STAC catalog metadata search filtering scenes with < 5% cloud cover.
- **11. Dataset Scale & Validation:** Testing on 1 unlabelled patch is anecdotal. Mitigation: Benchmark algorithms on public OSCD dataset and local sanity checks.
- **12. Misleading Confidence Scores:** Arbitrary formulas like '(1 - SSIM) * 100' are not probabilities. Mitigation: Label as 'Structural Divergence Metric'.
- **13. Visual Difference vs. Real Events:** Algorithms see reflectance changes, not building permits. Mitigation: Label outputs as 'Candidate Change Zones'.

THE FIVE FATAL CREDIBILITY KILLERS (INSTANT EXAMINATION FAILS)

1. The Cross-Sensor Comparison Flaw (Comparing Esri Wayback vs Google Satellite). 2. The Building Detection Fallacy (Claiming to detect single houses from 10m data). 3. Uncalibrated Area Math (Multiplying unreferenced pixels by an arbitrary slider). 4. Unmasked Cloud Flags (Highlighting cumulus clouds as detected construction). 5. Invented Accuracy Claims (Claiming '95% accuracy' without an audited test split).

OBSERVATION 7 Architectural Enhancement & Modular Pipeline Isolation

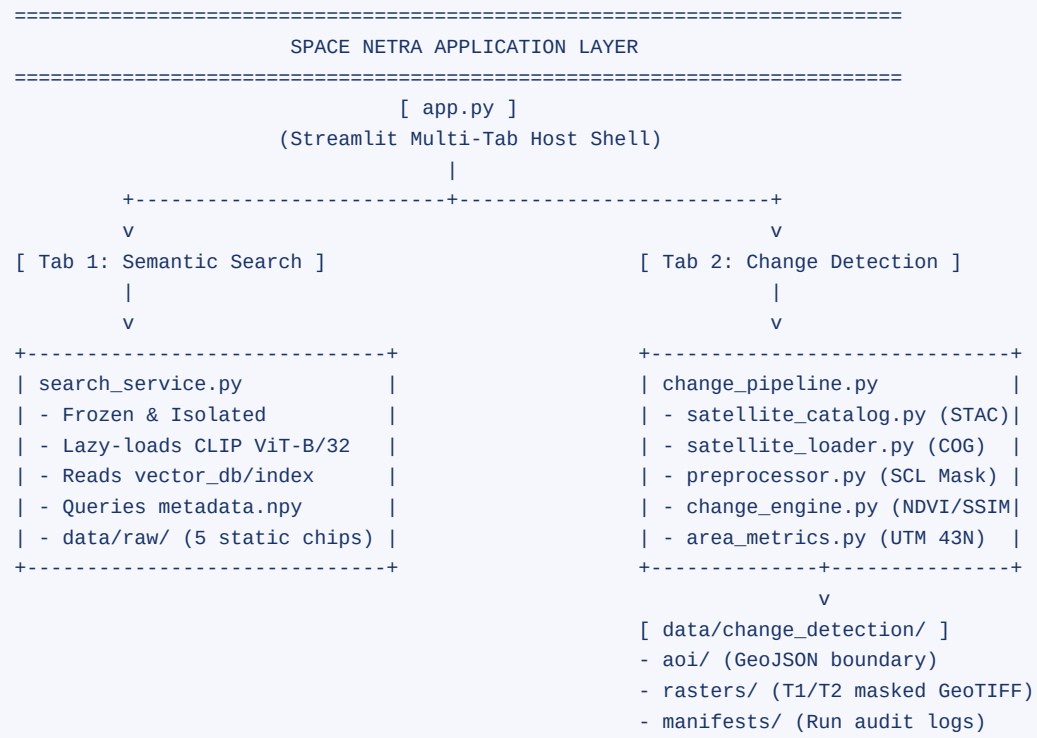
To introduce robust change detection without destabilizing working search capabilities, three architectural approaches were evaluated against software engineering principles.

Evaluation of Three Architectural Approaches

Architectural Approach	Complexity	Risk to Working Code	Student Viability	Recommendation
Approach A: Isolated Modular Pipelines	Low (Decoupled)	Near Zero (Safe)	Ideal (10/10)	RECOMMENDED
Approach B: Shared Intermediary Interface	High (Adapters)	Medium (Index updates)	Poor for V1 (4/10)	Defer to V2
Approach C: Full Geospatial Re-architecture	Very High (Rewrite)	Extreme (Destructive)	Unsuitable (1/10)	REJECTED

Target Modular Architecture (Approach A)

TARGET MODULAR SERVICE DECOUPLING



Component Isolation Boundaries

- **Strictly Independent:** vector_db/ and ingest.py must remain 100% isolated. Temporal GeoTIFFs must never be injected into the static FAISS index.
- **Shared UI Utilities:** The Streamlit navigation bar, Folium basemap viewer, Nominatim geocoder, and Plotly charting helpers are safely shared.

OBSERVATION 8

Master Engineering Roadmap & Measurable Success Criteria

A 5-phase engineering roadmap is established to guide implementation from clean environment setup to final demo freeze, coupled with strict, measurable acceptance gates.

Five-Phase Development Roadmap

Phase & Milestone	Core Engineering Tasks	Deliverable Artifacts
Phase 1: Asset Protection	Backup vector_db/; verify Python venv; lock 2	Backup archive, sambhajinagar_aoi.geojson
Phase 2: Ingestion & Prep	Build STAC client & COG loader; SCL cloud mas	satellite_catalog.py, preprocessor.py
Phase 3: Core Change Engine	Implement Delta-NDVI & SSIM differencing; UTM	change_engine.py, area_metrics.py
Phase 4: UI Integration	Mount Tab 2 in app.py; render Folium sync & P	Integrated app.py with limitations banner
Phase 5: Validation Freeze	Run zero-change sanity test; verify CLIP sear	Final demo freeze, technical report (.txt)

Pilot Use Case: Chhatrapati Sambhajinagar Expansion

- **Target Coordinates:** 2x2 km bounding box over the Shendra MIDC / DMIC industrial expansion corridor.
- **Temporal Observation Pair:** T1 = March 2021 (Baseline pre-monsoon) vs. T2 = March 2025/2026 (Recent pre-monsoon).
- **Detectable Target:** Macro-scale conversion of fallow agricultural plots into graded industrial building pads.

Six Measurable Acceptance Criteria for Version 1.0

- **Criterion 1 (Search Preservation):** Natural-language queries ('urban', 'forest') return valid top-k matches from vector_db/ without errors.
- **Criterion 2 (Real Satellite Ingestion):** System ingests authentic Sentinel-2 L2A GeoTIFFs via STAC rather than scraped web tiles.
- **Criterion 3 (Cloud Suppression):** Cumulus clouds in test scenes are masked out via the SCL layer and excluded from change metrics.
- **Criterion 4 (Sub-Pixel Coregistration):** Rasters align to identical UTM Zone 43N grid specs with < 0.5 px boundary jitter.
- **Criterion 5 (Defensible Area Math):** Reported change area in hectares is mathematically derived from pixel counts in UTM 43N (1 px = 0.01 ha).
- **Criterion 6 (Scientific Presentation):** The UI prominently displays sensing timestamps, platform metadata, and explicit 10m resolution caveats.

FINAL PLANNING RECOMMENDATION: PROCEED TO IMPLEMENTATION

With all 8 architectural and scientific observations formally documented, the project has eliminated theoretical ambiguities. Implementation can proceed with high confidence without risking existing vector search assets.